

Module 02: Agents in Crochet Circles

Learning Objectives

1. Describe yourself as a **free energy minimizing agent** when you sit down to crochet.
2. Explain how holding a pattern in mind functions as a **generative model** that guides your stitches.
3. Recognize **frogging** as a prediction error response — correcting when reality diverges from expectation.

Introduction

In the previous module, we identified the crochet project as a system with internal states, external states, and a Markov blanket at the working edge. But who is at the center of that system, making decisions, pulling yarn, and counting stitches? You are. In Active Inference, an **agent** is any entity that acts on its environment in ways that reduce surprise — that minimize the difference between what it expects and what it encounters. Every time you pick up your hook, you become that agent.

This module explores what it means to be a crocheter through the lens of Active Inference. You hold a pattern in your head (a generative model), you count stitches to confirm you are on track (belief updating), and when things go wrong, you rip back and try again (prediction error correction). These are not just craft skills — they are the core operations of an active inference agent.

Key Concepts

1. The Crocheter as Free Energy Minimizing Agent

In Active Inference, a **free energy minimizing agent** is one that continuously acts to reduce the discrepancy between its expectations (its model of the world) and its sensory observations. The agent does not passively receive information; it actively seeks states that confirm its predictions.

When you crochet, you are doing exactly this. You sit down with an expectation: "I will make a granny square" or "I will add six rows to this scarf." This expectation is your prior belief — your generative model predicting what the fabric should look like. As you work, you constantly compare what is emerging under your hands with what you expected. Is the row straight? Is the stitch count correct? Does the tension feel right?

When everything matches, you are in a low free energy state — things feel smooth, rhythmic, almost automatic. Experienced crocheters know this flow: the hook moves, the yarn feeds, the stitches accumulate, and the fabric grows exactly as planned. But when something goes wrong — a missed stitch, an unexpected bump in the fabric, a row that curves when it should be straight — free energy spikes. You feel that prickle of "something is off." This is surprise, in the technical Active Inference sense: the sensory evidence does not match your generative model's prediction.

The crocheter-as-agent has two fundamental options for reducing this free energy. First, **change the world to match the model**: rip back and redo the row (this is active inference in its most literal form — acting on the environment to make it conform to your expectations). Second, **change the model to match the world**: decide that the irregularity is acceptable, perhaps even desirable — "this is a design feature, not a mistake." Both strategies reduce free energy, but they do so in opposite directions.

2. The Pattern as Generative Model

A **generative model** in Active Inference is the agent's internal representation of how the world works — specifically, how sensory observations are generated by hidden causes. The model allows the agent to predict what it will observe before it observes it.

For a crocheter, the **pattern** is the most visible form of the generative model. Whether it is a written pattern from a book, a chart with symbols, or a design held entirely in your memory, the pattern tells you what to expect at every point in the project. "Chain 4, skip 2, double crochet in next stitch" — this is a prediction: if I perform these actions, I should see a specific configuration of stitches on my hook.

But the generative model is richer than the written pattern alone. It includes your knowledge of how yarn behaves — that worsted weight yarn will produce a denser fabric than lace weight

with the same hook, that acrylic has more stretch than cotton, that the first row after a foundation chain always feels tighter. It includes your understanding of stitch mechanics — that a yarn over pulled through two loops makes a single crochet, that skipping a stitch creates a hole. All of this knowledge, accumulated over your crafting life, forms the generative model you bring to every project.

An experienced crocheter's generative model is deeply refined. They can look at a finished garment and mentally decompose it into stitches, rows, and construction techniques. They can predict how a yarn will behave before they swatch. A beginner's model is sparser — they may follow a pattern stitch by stitch without understanding why each instruction leads to the intended result. Both are generative models; they differ in **depth** and **precision**.

3. Counting Stitches and Frogging: Belief Updating and Prediction Error

Belief updating is the process by which an agent revises its internal estimates based on new sensory evidence. In crochet, the most common form of belief updating is **stitch counting**. After completing a row, you count the stitches: "...twelve, thirteen, fourteen. Fourteen. Good — that matches." Your generative model predicted fourteen stitches; your observation confirms it. Your beliefs are updated with increased confidence.

But suppose you count and find only thirteen. Now there is a **prediction error**: the model said fourteen, but reality says thirteen. You have several options, each illustrating a different aspect of Active Inference:

- **Re-count** (refining the observation): Maybe you miscounted. You count again, more carefully. This is perceptual inference — adjusting your observation to see if the prediction error resolves.
- **Inspect the row** (searching for the cause): You look for a missed stitch, an accidental decrease. This is hypothesis testing — trying to identify which hidden cause (a specific error in a specific location) generated the surprising observation.
- **Frog the row** (corrective action): If you find the error, you rip back to the point of the mistake and redo the work. This is active inference at its most direct: changing the world (the fabric) to match the model (the pattern).

- **Accept the count** (model updating): Perhaps you decide thirteen is close enough, or you will add a stitch in the next row to compensate. This is adjusting the generative model — revising your expectations to accommodate the evidence.

Frogging deserves special attention. The word comes from "rip it, rip it" (like a frog's croak), and it is one of the crocheter's most powerful tools. Frogging is painful — it means undoing work you have already done. But it is also deeply rational from an Active Inference perspective. When accumulated prediction errors become too large (the piece is the wrong size, the pattern is visibly off, the gauge is completely wrong), the most efficient way to reduce free energy is to reset. Experienced crocheters learn when to frog and when to compensate — this wisdom is itself a form of refined generative model, a meta-model about when to act on the world and when to update beliefs.

Applications

In crochet, we see Agents manifest in:

- **Following a New Pattern for the First Time:** When you try a pattern you have never made before, you are operating with a sparse generative model. You rely heavily on the written instructions because your internal model cannot yet predict the outcome. Each row is a surprise — "Oh, that is what a front post double crochet looks like!" As you progress, your model fills in, and by the end of the project, you could likely make it again from memory. This is the agent's generative model being built through experience.
- **The "Tink" vs. "Frog" Decision:** "Tink" (knit spelled backward, borrowed into crochet culture) means undoing one stitch at a time. Frogging means ripping back many stitches or rows at once. The decision between tinking and frogging is a cost-benefit analysis performed by the agent: how much prediction error has accumulated, and what is the most efficient action to reduce it? Tinking corrects small, local errors (low prediction error, high precision correction). Frogging addresses systemic errors (high prediction error, wholesale reset). This decision-making process is a clear example of policy selection under Active Inference.

Conclusion

You, the crocheter, are an Active Inference agent. Your pattern is your generative model, your stitch counting is belief updating, and your frogging is prediction error correction. Every time you sit in your crochet circle and work a row, you are minimizing free energy — reducing the gap between what you expect to create and what your hands actually produce. In the next module, we will look more closely at the sensory side of this process: how you feel tension, detect errors, and perceive the fabric taking shape under your fingers.